members of the background population among whom *r,* the coefficient of relatedness, approaches zero. I could calculate *r* between me and the Queen using the one pathway for which records exist, but it would, as the definition requires, be so close to zero as to make no difference.

The reason for all that bewildering multiplicity of cousinship is sex. We have two parents, four grandparents, eight great grandparents, and so on, up to astronomical numbers. If you go on multiplying by two back to the time of William the Conqueror, the number of your ancestors (and mine, and the Queen's and the postman's) would be at least a billion, which is more than the world population at the time. That calculation alone proves that, wherever you come from, we share many of our ancestors (ultimately all if you go sufficiently far back) and are cousins of each other many times over.

All that complexity disappears if you look at cousinship from the gene's point of view (the point of view advocated, in different ways, throughout this book) as opposed to the individual organism's point of view (as has been conventional among biologists). Stop asking: What kind of cousin am I to my wife (the postman, the Queen)? Instead, ask the question from the point of view of a single gene, say my gene for blue eyes: What relation is my blue eye gene to the postman's blue eye gene? Polymorphisms such as the ABO blood groups go way back in history, and are shared by other apes and even monkeys. The A gene in a human sees the equivalent gene in a chimp as a closer cousin than the B gene in a human. As for the SRY gene on the Y chromosome, which determines maleness, my SRY gene 'looks upon' the SRY gene of a kangaroo as its kissing cousin.

Or we can look at relatedness from a mitochondrion's point of view. Mitochondria are tiny bodies teeming in all our cells, vital to our survival. They reproduce asexually and retain the remnants of their own genomes (they are remotely descended from free-living bacteria). By the Williams definition, a mitochondrial genome can be thought of as a single 'gene'. We get our mitochondria from our mothers only. So if we were now to ask how close is the cousinship of your mitochondria to the Queen's mitochondria there is a single answer. We may not know what that answer is, but we do know that her mitochondria and yours are cousins in only one way, not hundreds of ways as is the case from the point of view of the body as a whole. Trace your ancestry back through the generations, but always only through the maternal line and you